NewsBot 2.0

An End-to-End NLP Pipeline for Automated News Article Analysis

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From Midterm to Final

Midterm Foundation

- Text preprocessing pipeline
- TF-IDF vectorization
- Naive Bayes classification
- VADER sentiment analysis
- Named Entity Recognition
- Part-of-speech tagging

Final Enhancements

- LDA topic modeling
- Text summarization
- Multilingual support (EN/DE/FR/ES)
- Semantic search capabilities
- Conversational chatbot interface

The Problem

Volume Overload

Thousands of news articles published daily across multiple sources and categories

Manual Processing

Categorizing and analyzing each article manually consumes valuable time and resources

Fragmented Insights

Getting multiple perspectives requires running separate analyses one at a time

The Solution

One article input. Multiple insights output — all returned by a single function call

01

Classification

Category prediction with confidence scoring

02

Sentiment Analysis

Emotional tone detection

03

Topic Modeling

Hidden theme discovery

04

Summarization

Key sentence extraction

05

Entity Recognition

People, organizations, locations

06

Translation

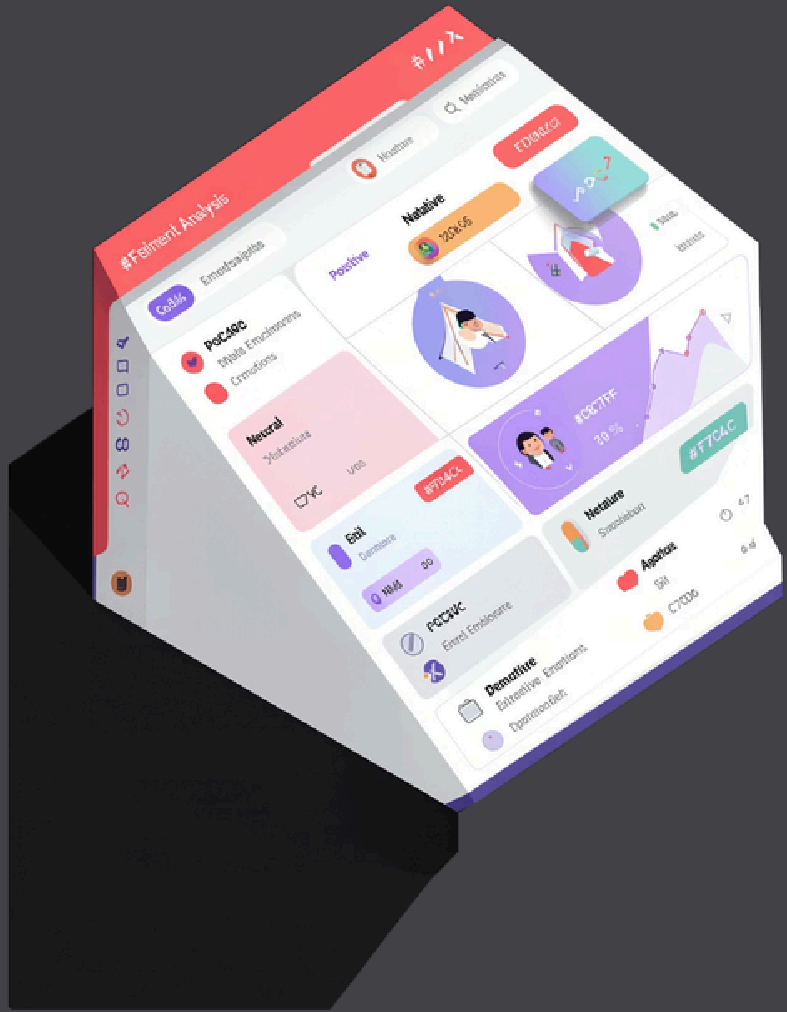
Multilingual support

07

Semantic Search

Relevant article retrieval

> `newsbot.analyze(article)` — A complete NLP pipeline in one function



Classification & Sentiment

Category Classification

Trained both Naive Bayes and SVM models, with automatic selection of the best performer. Returns predicted category with confidence score across 5 classes.

- Business
- Sport
- Politics
- Tech
- Entertainment

Sentiment Analysis

VADER scores each article using compound scoring, returning positive/negative/neutral labels with detailed intensity metrics.

Performance: Classification accuracy exceeds 96% on test data

Topic Modeling & Summarization

1	**LDA Topic Discovery** Latent Dirichlet Allocation uncovered 5 hidden themes across 1,490 articles without labeled training data • Top 10 words per topic identified • Article distribution analyzed • No manual labeling required
2	**LSA Summarization** Latent Semantic Analysis from sumy library extracts 3 most important sentences • Generates concise headline • Calculates compression ratio • Maintains core meaning

Entities, Multilingual & Search

Named Entity Recognition

spaCy efficiently extracts people, organizations, and geographic locations from each article with high precision

Multilingual Translation

langdetect identifies input language, MyMemory translator converts to German, French, or Spanish

Semantic Search

TF-IDF vectorization combined with cosine similarity finds top 5 most relevant articles for any query

> *"The bank reported record profits" → "Die Bank meldete Rekordgewinne"*

Demo Results

Sample output from `newsbot.analyze()` function demonstrating complete pipeline capabilities

Category	Sentiment	Language
Business · Confidence: High	**Negative** · Score: −0.97	**English** (en)

Key Entities Identified

People

- Arthur Andersen
- Cynthia Cooper

Locations

- United States
- New York

Generated Summary

Three most important sentences automatically extracted using LSA algorithm, capturing core article meaning with high compression ratio.

Limitations

Model Architecture

Uses classical ML approaches (Naive Bayes, SVM) rather than deep learning architectures like BERT or transformers

Chatbot Scope

Rule-based conversational interface with predefined responses, not fully dynamic or context-aware

Training Data

Trained exclusively on English BBC articles, limiting cross-lingual classification accuracy

Translation Constraints

MyMemory API imposes 500-character limit, truncating longer articles during translation

Conclusion

Complete Pipeline

Built end-to-end NLP system processing any news article through 7 distinct analysis modules

Four New Modules

Expanded significantly beyond midterm with topic modeling, summarization, multilingual support, and semantic search

High Accuracy

Achieved classification accuracy exceeding 96% on test dataset

Integrated Workflow

Demonstrated how multiple NLP techniques integrate into one practical, automated solution

Impact: Reduces manual analysis time from minutes to seconds while providing comprehensive insights. Human reviewers still validate output, but the heavy computational lifting is automated.